

**BINAURAL AUDIO**

Binaural audio is considered to provide a really natural sounding three dimensional sound. It's recorded by placing two stereo microphones in ear-like cavities inside a dummy head, thus, helping capture and recreate the sound almost exactly how you'll be hearing it.  
<http://dgit.in/3DBinAudio>

**DRONE RAGE**

What were initially considered to be remotely controlled flying objects for military operations have now become something that everyone either wants or hates. Drones have crept into diverse fields, be it for hobby, defence or even for commercial purposes so you should read more about these sky demons.  
<http://dgit.in/Droneland>

**RISE OF RANSOMWARE**

Ransomware is a type of malware that locks certain important parts of your PC like the keyboard for example and holds your personal data for ransom by not letting you access it. There has been a sudden rise in ransomware recently as hackers have started using ransom cryptware that goes a level above using encryption.  
<http://dgit.in/RansomLink>

**HYPERSONIC AIRLINERS?**

Concorde and Tupolev Tu-144 have known to be the only two civilian airliners that were able to break the speed of sound but both of these have been decommissioned since long. That hasn't been the end for supersonic airliners as researchers are creating faster ones.  
[http://dgit.in/Hyper\\_Sonic](http://dgit.in/Hyper_Sonic)

SHORTS

# With hardware saturation, let the software wars begin!

Be it the Apple A9 or Qualcomm's Snapdragon 820, or even camera sensors etc, hardware is no longer the difference maker in consumer tech.

– By Prasad Banerjee

Look down at your phone? Do you see those icons that look different from what they do on your friend's phone? Good. Now go and find out the list of sensors on the fitness wearable you really like and compare it to your second favourite. What do you see? They're the same right? Except in the app that comes with them. That's exactly what we're talking about here.

Qualcomm says its Snapdragon 820 is X% more powerful than older generations, as does Apple about its A9. The fact though is that hardware is redundant



do what is needed on the software side. The best example of this in fact is Apple, which for a long time remained unparalleled in camera performance, despite a measly 8 megapixel sensor. It is also one of the reasons why megapixel count doesn't determine camera quality over a certain threshold.



today. Phones in a particular price bracket are essentially the same, separated only by their software gimmicks. There's much more to this though. In the world of incremental innovation, we stand at a point where the race for the best software technology is more important than ever.

**From UI to OS**

UIs have for the longest time been seen as measures taken by companies to differentiate their product from others, but it's not how good the MiUI looks that gives Xiaomi the edge. It's actually the processes in the background. There's a reason things like MiUI, Nubia etc. are referred to as OSs. It's because they're no longer cosmetic changes. They actually manage resources differently or add something else in the background. Here's how.

**Camera**

Ever wondered why the Moto Turbo, despite its 21MP sensor has a sub-par camera? It's because Google's stock camera app isn't good. It's simple and easy to use, but companies like Sony, Samsung, Xiaomi, LG, ZTE etc. have put in tons of research to

**Memory Management**

Try using a Xiaomi phone running MiUI alongside a Motorola phone with its stock Android interface. You'll see that with no apps running in the background both phones seem to consume the same amount of RAM. But how can that be?

It's because Xiaomi replaces instead of layering. It takes the open source Android code and replaces various parts of that code, while keeping the base the same. As a result, MiUI doesn't consume extra memory. The stock Android camera app is not present on a Xiaomi phone and has been replaced by the Mi Camera app. The same is done by most other companies.

That should still not account for all the extra features Xiaomi provides though. And that is where MIUI's memory management comes in. It's what differentiates one